

Q1 Handling ArithmeticException

10 marks

Write a Java program to divide two integers entered by the user. If the denominator is zero, handle the exception using a try-catch block and display an appropriate error message.

Your Code

```
1 import java.util.*;
2 class Arithmetic{
3
4     static int divide(int a,int b){
5         return a/b;
6     }
7     static void display(int x){
8         System.out.println("result: "+x);
9     }
10    public static void main(String[] args){
11        Scanner sc=new Scanner(System.in);
12        int a=sc.nextInt();
13        int b=sc.nextInt();
14        try{
15            Arithmetic.display(Arithmetic.divide(a,b));
16        }
17        catch(ArithmeticException e){
18            System.out.println("cannot divide by zero");
19        }
20        finally{
21            sc.close();
22        }
23    }
24 }
```

Input

20
4

Output

result: 5

Q2 Handling ArrayIndexOutOfBoundsException

10 marks

Write a Java program to create an integer array of five elements. Ask the user to enter an index and display the corresponding element. If the index is outside the valid range, handle the exception and display an error message.

Your Code

```
1 import java.util.*;
2 class array{
3     static int a[]={1,2,3,4,5};
4     static void display(int n){
5         System.out.println("element: "+a[n]);
6     }
7     public static void main(String[] args){
8         Scanner sc=new Scanner(System.in);
9         int n=sc.nextInt();
10        try{
11            display(n);
12        }
13        catch(ArrayIndexOutOfBoundsException e){
14            System.out.println("invalid index");
15        }
16        finally{
17            sc.close();
18        }
19    }
20 }
```

Input

10

Output

invalid index

Run

Save

Q3 Handling NumberFormatException

10 marks

Write a Java program to accept a number as a string and convert it into an integer using Integer.parseInt(). If the input is not a valid integer, handle the exception and display an appropriate message.

Your Code

```
1 import java.util.*;
2 public class main{
3     public static void main(String[] args){
4         Scanner sc=new Scanner(System.in);
5         try{
6             String s=sc.nextLine();
7             int n=Integer.parseInt(s);
8             System.out.println(n);
9         }
10        catch(NumberFormatException e){
11            System.out.println("invalid input");
12        }
13        finally{
14            System.out.println("program executed");
15        }
16        sc.close();
17    }
18 }
```

Run

Save

Input

546

Output

```
546
program executed
```

Q4 Handling Multiple Exceptions

10 marks

Write a Java program that performs the following operations:

Accept two integers from the user.

Divide the first number by the second.

Store the result in an array.

Handle both ArithmeticException and ArrayIndexOutOfBoundsException using multiple catch blocks.

Your Code

```
1 import java.util.*;
2 public class main{
3     public static void main(String[] args){
4         Scanner sc=new Scanner(System.in);
5         int []arr=new int[5];
6         try{
7             int a=sc.nextInt();
8             int b=sc.nextInt();
9             int res=a/b;
10            int index=sc.nextInt();
11            arr[index]=res;
12            System.out.println(arr[index]);
13        }
14        catch(ArithmeticException e){
15            System.out.println("cannot divide by zero");
16        }
17        catch(ArrayIndexOutOfBoundsException e){
18            System.out.println("invalid index enter from(0-4)");
19        }
20        finally{
21            System.out.println("program executed");
22        }
23        sc.close();
24    }
25 }
```

Input

10
1
10

Output

```
invalid index enter from(0-4)
program executed
```